

CSE400 – Lecture 3 Scribe

Project Kickoff and Undergraduate Research Programme (UGRP)

1. Course Project Component

1.1 Definition

The CSE400 course includes a **Project Kickoff** component carrying a weightage of **30%** of the total course evaluation.

1.2 Assumption

Since the project carries significant weightage, it is intended to be a core learning and evaluation mechanism rather than an auxiliary activity.

2. Project Team Formation

2.1 Requirement

Deadline: 17 January 2026 (Saturday, End of Day)

2.2 Reasoning

- The project is group-based.
- Group-based execution requires early coordination.
- A fixed team formation deadline ensures timely initiation of project milestones.

3. Project Execution Guidelines

3.1 Weightage

Total project weightage: **30%**

3.2 Structure

- The project consists of six major milestones (M1–M6).
- One submission per group is required for each milestone.

3.3 Assumption

The milestones are sequential and cumulative, meaning each stage builds upon the previous one.

4. Milestone-Wise Project Execution

4.1 M1: Kickstart

4.1.1 Components

- Team Formation
- Area Identification
- Background
- Motivation
- Problem Formulation

4.1.2 Step-by-Step Reasoning

- A project must begin with a clearly identified domain.
- Understanding the background provides context.
- Motivation justifies the importance of the problem.
- Based on background and motivation, the problem is formally formulated.

4.2 M2: Mathematical Modeling

4.2.1 Definition

Mathematical modeling of the selected problem using:

- Random Variables (RV)
- PMF / PDF
- CDF
- Multivariate Random Variables
- Joint PMF / PDF / CDF

4.2.2 Step-by-Step Reasoning

- Real-world problems contain uncertainty.
- Uncertainty is modeled using random variables.
- PMF/PDF describes probabilistic behavior.
- CDF captures cumulative probability.
- Joint distributions are required for multiple variables.

4.3 M3: Coding

4.3.1 Components

- Simulation
- Computation

4.3.2 Reasoning

- Mathematical models must be validated.
- Simulation reproduces model behavior.
- Computation provides numerical results.

4.4 M4: Inference

4.4.1 Components

- Choose a randomized algorithm
- Understand the algorithm
- Code the algorithm

4.4.2 Assumption

Randomized algorithms are suitable for inference in the modeled problem domain.

4.4.3 Reasoning

- Algorithm selection precedes implementation.
- Understanding ensures correctness.
- Coding enables evaluation.

4.5 M5: Randomized Algorithms

4.5.1 Components

- Apply the randomized algorithm
- Generate results
- Compare with deterministic algorithms

4.5.2 Step-by-Step Reasoning

- Apply the algorithm to the problem.
- Obtain execution results.
- Compare with deterministic methods.
- Draw new inferences.

4.6 M6: Bounds and Analysis

4.6.1 Components

- Derive bounds
- Perform analysis

- Compile final deliverables

4.6.2 Reasoning

- Theoretical bounds validate performance.
- Analysis supports empirical results.
- All outputs are compiled for final submission.

5. Project Deliverables and Assessment

5.1 Deliverables

- Codes
- Reports
- Videos
- Other milestone-specific artifacts

5.2 Assessment

- Team assessment before midsemester
- Team assessment after midsemester
- Final evaluation:
 - Project Viva
 - Final Submission

6. Submission #1: Concept Evolution Maps

6.1 Definition

Concept evolution maps visualize how project ideas evolve from initial concepts to final problem formulation.

6.2 Tools

- Miro
- draw.io (diagrams.net)

7. Submission #2: Scribe – Learning Reflection Logs

7.1 Types

- Lecture Scribe
- Project Scribe

7.2 Lecture Scribe

- Assigned to two groups per lecture
- Reflects lecture content
- Minimum length: 8–10 pages

7.3 Project Scribe Components

- Decision logs (Why X over Y?)
- Constraints and alternatives
- Final decisions
- Evidence and justification
- Trade-off matrices (Cost vs Performance vs Risk)

8. Submission #3: Multimodal Artifacts

8.1 Definition

Artifacts may include video, audio, or visual materials.

8.2 Requirements

- One video per milestone
- Duration: 10–15 minutes
- Focus on content, not editing

8.3 Types

- Think-Aloud Videos
- One-Minute Insight Videos
- Project Demo

9. Introduction to UGRP-8 (2026–2027)

UGRP is introduced as a structured undergraduate research initiative.

10. Motivation for UGRP

10.1 T-Shaped Engineer Concept

Vertical bar: Depth in a single technical discipline

Horizontal bar: Ability to apply knowledge across disciplines and collaborate

10.2 Reasoning

UGRP aims to develop both technical depth and interdisciplinary breadth.

11. Philosophy of UGRP

11.1 Core Principles

- Multidisciplinary
- Experiential Learning
- Research Driven

11.2 4D Model

1. Discover
2. Design
3. Develop
4. Deliver

12. Technical Breadth Under UGRP

Includes:

- CS and CSE
- Data Science and Applied AI
- Modern Computer System Design
- Networks
- IoT / IoBNT / IoV

13. Industry and Research Activities

13.1 Example

5G-enabled Intelligent Transportation Systems Testbed in Gujarat

13.2 Activities

- V2X communication
- Sensor data collection
- ITS dashboard development
- Deep learning and deep-RL solutions
- mmWave beam prediction
- C-V2X and Wi-Fi coexistence

14. Collaborations and Outcomes

14.1 Evidence Presented

- International academic collaborations
- IEEE conference and journal publications
- Awards and recognitions
- Alumni placements in industry and academia

15. Conclusion of Lecture 3

Lecture 3 establishes:

- The project execution framework of CSE400
- A milestone-driven research methodology
- The UGRP philosophy and its academic and industrial outcomes

The lecture emphasizes process, structure, and research orientation, forming the foundation for subsequent technical work in the course.